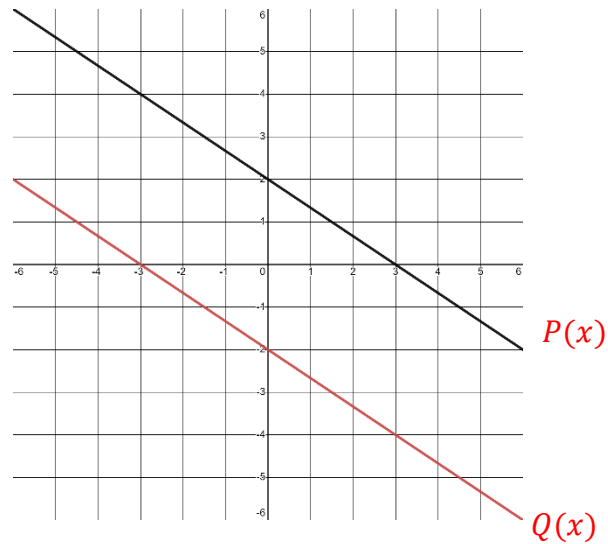


The graph of the function $P(x)$ is shown.



1. Write the function, $P(x)$, that represents the line.

$$P(x) = -\frac{2}{3}x + 2$$

2. Draw a line that is parallel to $P(x)$ with a y -intercept of -2 . Label this line $Q(x)$.
3. Describe how $P(x)$ can be shifted to coincide with $Q(x)$.

Shift line down 4.

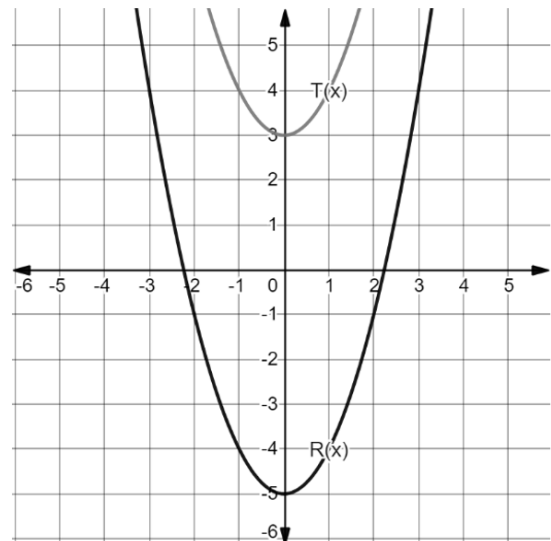
4. Write the function $Q(x)$ in terms of $P(x)$.

$$Q(x) = P(x) - 4$$

5. The graphs of $R(x)$ and $T(x)$ are shown on the graph.

Write the transformation by completing the equation.

$$T(x) = R(x) + 8$$



The functions $h(x)$ and $j(x)$ are shown on the graph.

6. Write $h(x)$ in vertex form.

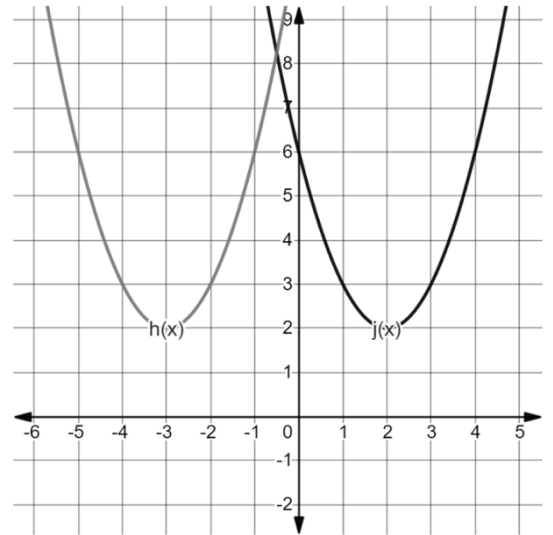
$$h(x) = (x + 3)^2 + 2$$

7. Write $j(x)$ in vertex form.

$$j(x) = (x - 2)^2 + 2$$

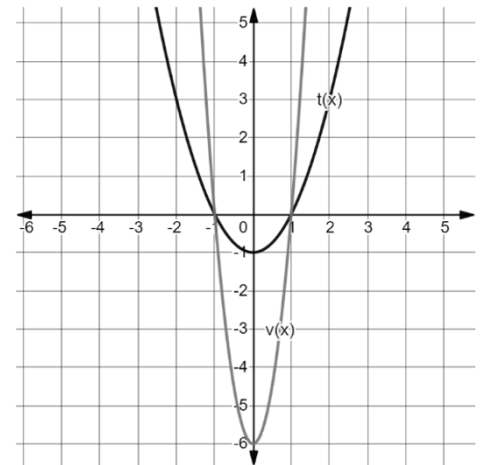
8. Describe how $h(x)$ was transformed to create $j(x)$.

$h(x)$ shifted 5 units right



9. The functions $t(x)$ and $v(x)$ are shown on the graph, where $v(x)$ is a transformation of $t(x)$. What is the value of k such that $v(x) = kt(x)$?

$$k = 6$$



10. The table of values for $w(x)$ and $z(x)$ are shown, where $z(x)$ is a transformation of $w(x)$. Write a description of the transformation.

$y = w(x)$		$y = z(x)$	
x	y	x	y
-2	8	-2	13
-1	2	-1	7
0	0	0	5
1	2	1	7
2	8	2	13

$w(x)$ shifted up 5 units which becomes $z(x)$.